

# User Study – Manual Python Workflow (Method 1 Only)

*Instructions:* Fill Section A once. For Section B, answer *per use-case*. Duplicate Section B if you completed multiple use-cases.

Participant ID: P8

— **Your assignment:** ☒ **Method 1** (Manual Python)  
River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

## Section A – Background (answer once)

### A1. Educational level (*highest completed*)

☐ High School ☐ Vocational/Technical ☒ Bachelor's ☐ Master's ☐ Doctorate ☐ Other:

### A2. Primary role (*pick the closest*)

☐ IoT/CPS Developer ☒ Data Scientist / ML Engineer ☒ Embedded SW Engineer ☐  
Educator / Researcher ☐ Software Architect / Full-stack ☐ System Integrator ☐ Other:

### A3. Self-rated expertise (*tick one per row*)

|                                  | Beginner                 | Intermediate                        | Advanced                            | Expert                   |
|----------------------------------|--------------------------|-------------------------------------|-------------------------------------|--------------------------|
| Python programming               | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Time-series / ML                 | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Domain knowledge (your use-case) | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |

### A4. Prior experience (*used ML frameworks/libraries before?*) ☒ Yes ☐ No

## Section B – Task-Specific (duplicate this page *per use-case*)

This B-section is for: ☐ U1 River Flow ☒ U2 Smart-Home Energy ☐ U3 Solar Energy

### B1. Time & Effort

*Record your own times. Estimate if needed. “Extra learning time” = time spent reading docs/tutorials.*

#### B1. Time spent (minutes)

|                                   | Method 1 (Manual Python) |
|-----------------------------------|--------------------------|
| Writing/testing code (end-to-end) | 6                        |
| Extra learning time               | 1                        |

### B2. Lines of code (LoC) (*exclude blanks/comments*)

Manual Python LoC: 500 Lines

## B2. Task Completion & Ease of Use

Completion: None = could not finish; Partial = finished some parts; Complete = all parts.

### B3. Were you able to finish the use-case with Manual Python?

Method 1 (Manual Python): ☐ None ☐ Partial ☒ Complete

### B4. Ease by workflow step (Manual Python) (1=Very easy, 5=Very difficult; tick one per cell)

| Step                               | 1                        | 2                                   | 3                                   | 4                                   | 5                        |
|------------------------------------|--------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------|
| Whole workflow                     | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |
| Data preprocessing                 | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> |
| Feature engineering (lags/rolling) | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> |
| Model training                     | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Model evaluation/plots             | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |

### B1. Which steps did you complete? (check all that apply)

- ☒ Preprocessing
- ☒ Model configuration
- ☒ Training
- ☒ Evaluation
- ☒ Plotting/Diagnostics

### B2. How difficult was each step? (1=Very easy, 5=Very difficult; tick one per row)

| Step                 | 1                        | 2                                   | 3                                   | 4                                   | 5                        |
|----------------------|--------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------|
| Preprocessing        | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> |
| Model configuration  | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Training             | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Evaluation           | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |
| Plotting/Diagnostics | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |

### B3. Notes (issues, decisions, or gaps to revisit):

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### B5. Time spent per step (record for this use-case; minutes or start/end)

| Step                                 | Start (hh:mm) | End (hh:mm)  | Minutes    |
|--------------------------------------|---------------|--------------|------------|
| Preprocessing                        | <u>10:00</u>  | <u>11:00</u> | <u>60</u>  |
| Model configuration                  | <u>11:00</u>  | <u>13:00</u> | <u>120</u> |
| Training                             | <u>14:00</u>  | <u>15:00</u> | <u>60</u>  |
| Evaluation (metrics/baseline)        | <u>15:00</u>  | <u>16:00</u> | <u>60</u>  |
| Plotting / Diagnostics               | <u>16:00</u>  | <u>16:30</u> | <u>30</u>  |
| Debugging (errors, crashes, outputs) | <u>16:30</u>  | <u>17:00</u> | <u>30</u>  |
| Extra learning / searching / docs    | <u>9:00</u>   | <u>10:00</u> | <u>60</u>  |
| Total                                |               |              | <u>420</u> |

*Notes:* “Learning” includes reading docs, web searches, tutorials, setup notes, and guides. “Debugging” covers error fixing, crashes, failed runs, and investigating unexpected results.

**B6. Comments (bugs, friction points, ideas):**

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